THE NOVA RESEARCH CURRICULUM

Biology

Grade 8 · Term 1

Student edition

Outcome: BI-8.1 variation, adaptation, genes, chromosomes, DNA, mitosis and meiosis.

Lesson notes

- **Variation and adaptation (Weeks 1–4):** continuous variation (height, mass) vs discontinuous (blood group, tongue rolling); caused by genes, environment or both. Adaptation: features that help survival — camel (stores fat, wide feet), cactus (spines, thick stem), markhor (climbing hooves in the mountains). Class height survey and histogram.
- **Genes, chromosomes and DNA (Weeks 5–8):** the nucleus holds chromosomes; human body cells have 46 (23 pairs). Chromosomes are made of DNA, a double helix; a gene is a section of DNA that controls a feature.

  ▶ **Try it on NovaSchool:** Zooming in: from cell to DNA — open **Simulations** on your dashboard.
- **Mitosis and meiosis (Weeks 9–12):** mitosis makes two identical cells for growth and repair. Meiosis makes four sex cells (gametes) with half the chromosomes (23), so that each child gets a mix from both parents.

  ▶ **Try it on NovaSchool:** Cell division: mitosis and meiosis — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1	Height survey histogram; continuous or discontinuous?
2	Adaptation cards for five Pakistani animals and plants.
3	Label nucleus, chromosome, DNA, gene.
4	Mitosis vs meiosis comparison table.

Question bank

#	Outcome	Question	Your answer
1	BI-8.1	Height is an example of ... variation.	discontinuous / continuous / no / chemical
2	BI-8.1	Blood group is an example of ... variation.	discontinuous / continuous / environmental only / none
3	BI-8.1	How many chromosomes are in a human body cell?	23 / 46 / 48 / 92
4	BI-8.1	A gene is a section of ...	protein / DNA / cytoplasm / cell wall

#	Outcome	Question	Your answer
5	BI-8.1	Mitosis produces …	four different cells / two identical cells / sex cells only / no cells
6	BI-8.1	Meiosis makes cells with half the number of chromosomes. True or false.	True / False
7	BI-8.1	Give one adaptation of a camel and explain it.	_____
8	BI-8.1	DNA has the shape of a …	single line / circle / double helix / square